Learning with AI

Spell Arena - Building Assitive Gaming Features

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ASE 485 – Capstone Project

How I Used AI

- Art direction
- Learning methods of implementation
- Inspiration over direct implementation
- Learning through communication
- Building Assitive Features

The Models I Used

- ChatGPT Free Tier (Discussions about art mood boards)
- Claude Sonnet (Code Review and Questioning)
- Cursor (Starting point for use of USS and UXML)

What I've Learned

- Communicating with AI helped me push through design roadblocks
- Art direction should be guided by humans, not generated entirely by AI
- UI Toolkit simplified UI development compared to previous methods

Forms of UI Tools in Unity

Unity GUI

- Built directly in the scene hierarchy using GameObjects
- Layout controlled with anchors, pivots, and layout groups
- Drag-and-drop workflow tied to the scene

Forms of UI Tools in Unity

UI Toolkit

- Inspired by web development (UXML + USS)
- UI defined in documents instead of GameObjects
- Uses retained-mode UI (better performance)
- Supports reusable styles and themes

Assistive Feature: Pinned Spells

- Players can pin frequently used spells to the UI
- Keeps important abilities visible at all times
- Reduces cognitive load during combat
- Helps players learn spell combinations faster

How Pinned Spells Work

- Player selects spells to pin
- Displayed in a fixed location on screen
- Prioritized visually over other UI elements
- Inspired by hotbar systems in other games

UI Implementation (UI Toolkit)

- Built with UXML (structure) and USS (styling)

Design Choices:

- Transparent background for immersion
- High-contrast icons for readability
- Consistent placement across screens
- Hover feedback for interaction clarity

Using UI to Teach Assistive Features

- UI acts as a learning tool, not just display
- Players learn through interaction

Techniques Used:

- Highlighted UI elements
- Subtle animations to draw attention
- Consistent icon design language

Art Direction Tips from AI

- Helped build mood boards and design direction
- Guided early environment ideas
- Influenced campaign scene design
- Reinforced importance of consistent themes

AI's Role in Design Decisions

- Helped reduce UI complexity
- Suggested accessibility-focused layouts
- Reinforced visual hierarchy and clarity
- Used for inspiration, not final assets

Impact on Player Experience

- Reduces frustration for new players
- Helps players focus on gameplay
- Encourages experimentation
- Makes the game more accessible

Conclusion

- AI is a tool to support creativity, not replace it
- Strong UI design improves player experience
- AI accelerated learning of UI Toolkit (UXML & USS)